Week 4 — Working Solver Output (Minimum Working Version)

Goal

Implement the solver for at least one basic case end-to-end.

Required Outputs

☐ Solver works for at least 1 basic case (numeric or symbolic)

☐ At least 3 valid test cases documented (expected vs actual)

☐ Trail shows real computation steps (not only the final answer)

Evidence to Attach

☐ Screenshot: valid run with full trail and final answer

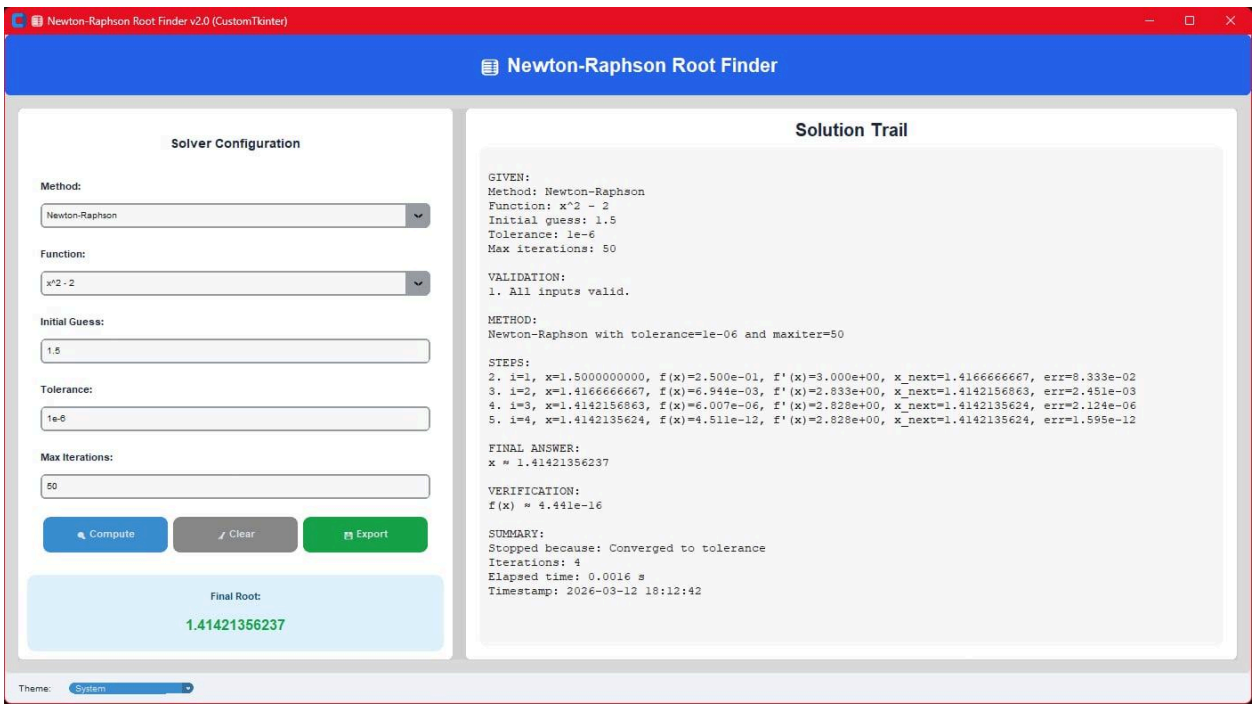
Reflection (Write short answers)

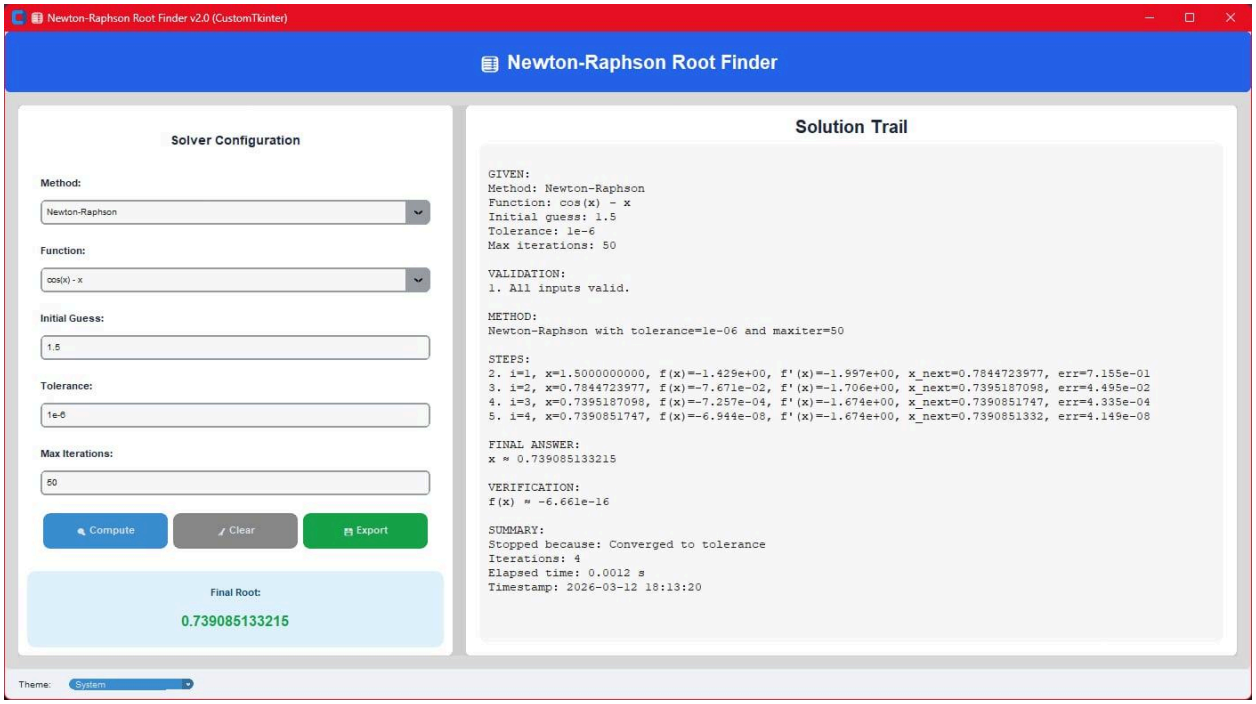
- What part of the solver is still incomplete?

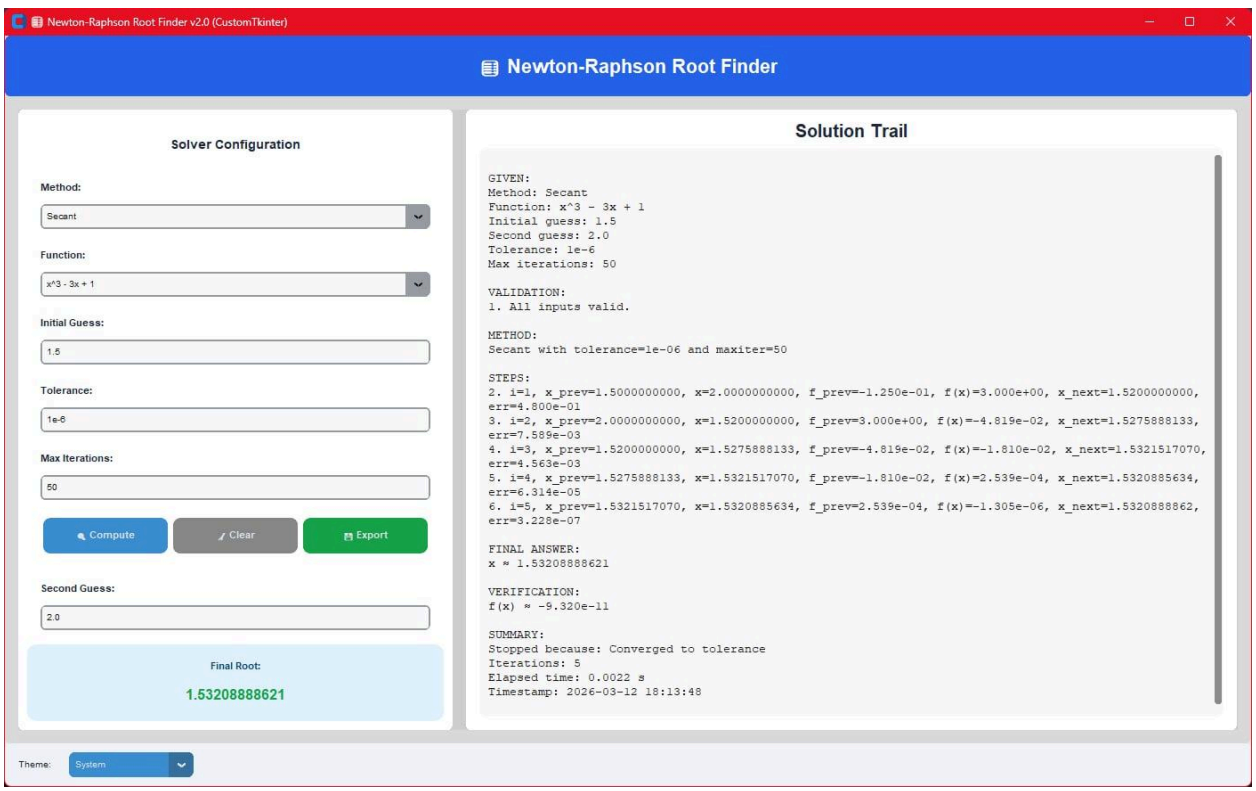
Test Case/ Evidence Output

Newton Raphson Method

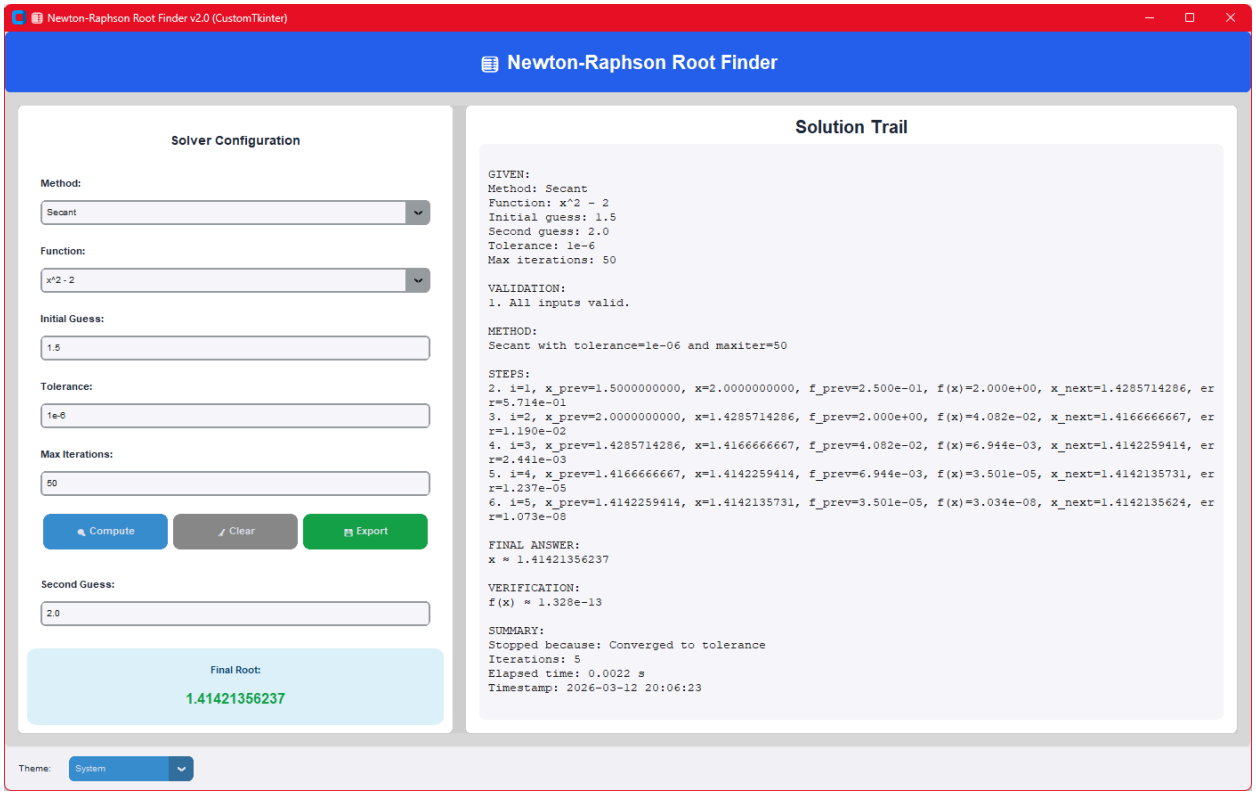
Test Case	Function	Initial Guess	Expected Root	Actual Root	Status
1	$x^2 - 2$	1.5	1.414213562	1.414213562	**Passed**
2	$\cos(x) - x$	0.5	0.739085	0.739085	**Passed**
3	$x^3 - 3x + 1$	0	0.347296	0.347296	**Passed**

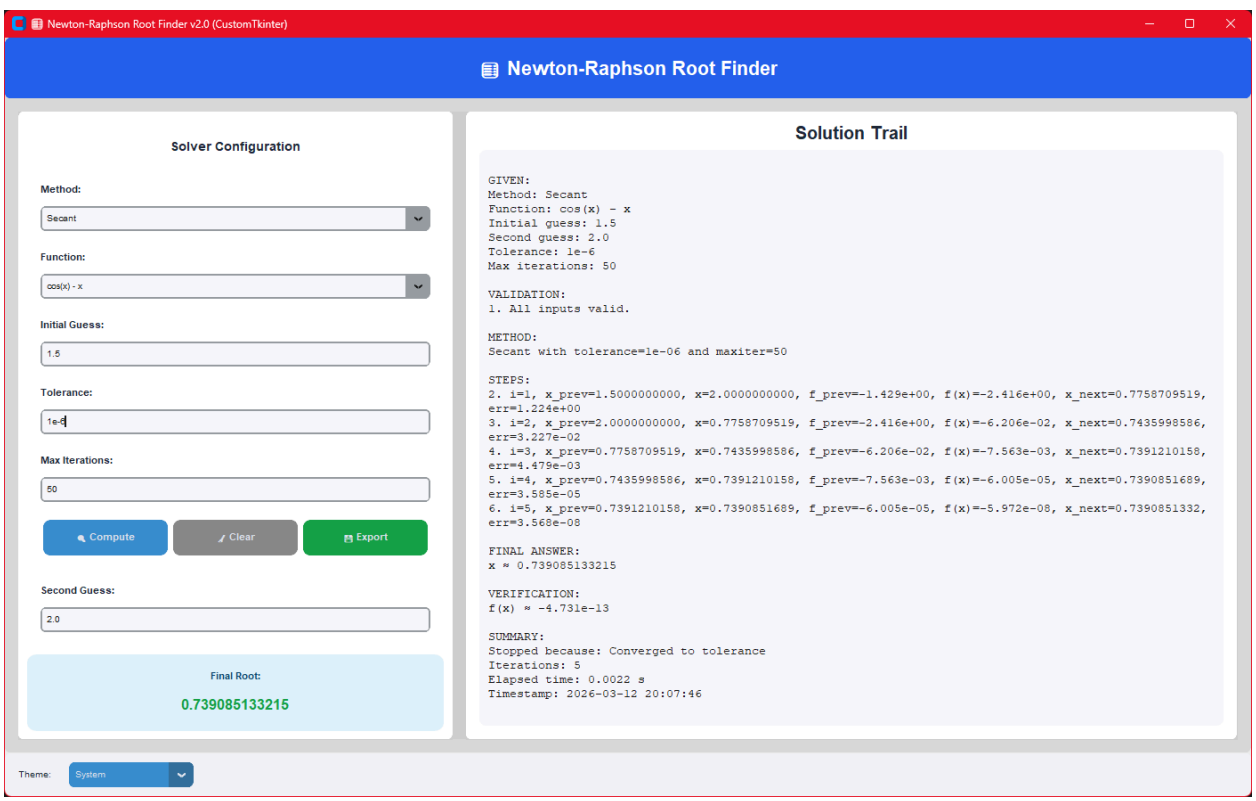



Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Newton-Raphson
Function:
cos(x) - x
Initial Guess:
1.5
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Final Root:
0.739085133215
Solution Trail
GIVEN:
Method: Newton-Raphson
Function: cos(x) - x
Initial guess: 1.5
Tolerance: 1e-6
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Newton-Raphson with tolerance=1e-06 and maxiter=50
STEPS:
2. i=1, x=1.5000000000, f(x)=-1.429e+00, f'(x)=-1.997e+00, x_next=0.7844723977, err=7.155e-01
3. i=2, x=0.7844723977, f(x)=-7.671e-02, f'(x)=-1.706e+00, x_next=0.7395187098, err=4.495e-02
4. i=3, x=0.7395187098, f(x)=-7.257e-04, f'(x)=-1.674e+00, x_next=0.7390851747, err=4.335e-04
5. i=4, x=0.7390851747, f(x)=-6.944e-08, f'(x)=-1.674e+00, x_next=0.7390851332, err=4.149e-08
FINAL ANSWER:
x ≈ 0.739085133215
VERIFICATION:
f(x) ≈ -6.661e-16
SUMMARY:
Stopped because: Converged to tolerance
Iterations: 4
Elapsed time: 0.0012 s
Timestamp: 2026-03-12 18:13:20
Theme:
System


Newton-Raphson Root Finder v2.0 (CustomTkinter)
Newton-Raphson Root Finder
Solver Configuration
Method:
Secant
Function:
x^3 - 3x + 1
Initial Guess:
1.5
Tolerance:
1e-6
Max Iterations:
50
Compute
Clear
Export
Second Guess:
2.0
Final Root:
1.53208888621
Solution Trail
GIVEN:
Method: Secant
Function: x^3 - 3x + 1
Initial guess: 1.5
Second guess: 2.0
Tolerance: 1e-6
Max iterations: 50
VALIDATION:
1. All inputs valid.
METHOD:
Secant with tolerance=1e-06 and maxiter=50
STEPS:
2. i=1, x_prev=1.5000000000, x=2.0000000000, f_prev=-1.250e-01, f(x)=3.000e+00, x_next=1.5200000000, err=4.800e-01
3. i=2, x_prev=2.0000000000, x=1.5200000000, f_prev=3.000e+00, f(x)=-4.819e-02, x_next=1.5275888133, err=7.589e-03
4. i=3, x_prev=1.5200000000, x=1.5275888133, f_prev=-4.819e-02, f(x)=-1.810e-02, x_next=1.5321517070, err=4.563e-03
5. i=4, x_prev=1.5275888133, x=1.5321517070, f_prev=-1.810e-02, f(x)=2.539e-04, x_next=1.5320885634, err=6.314e-05
6. i=5, x_prev=1.5321517070, x=1.5320885634, f_prev=2.539e-04, f(x)=-1.305e-06, x_next=1.5320888862, err=3.228e-07
FINAL ANSWER:
x ≈ 1.53208888621
VERIFICATION:
f(x) ≈ -9.320e-11
SUMMARY:
Stopped because: Converged to tolerance
Iterations: 5
Elapsed time: 0.0022 s
Timestamp: 2026-03-12 18:13:48
Theme:
System

Secant Method





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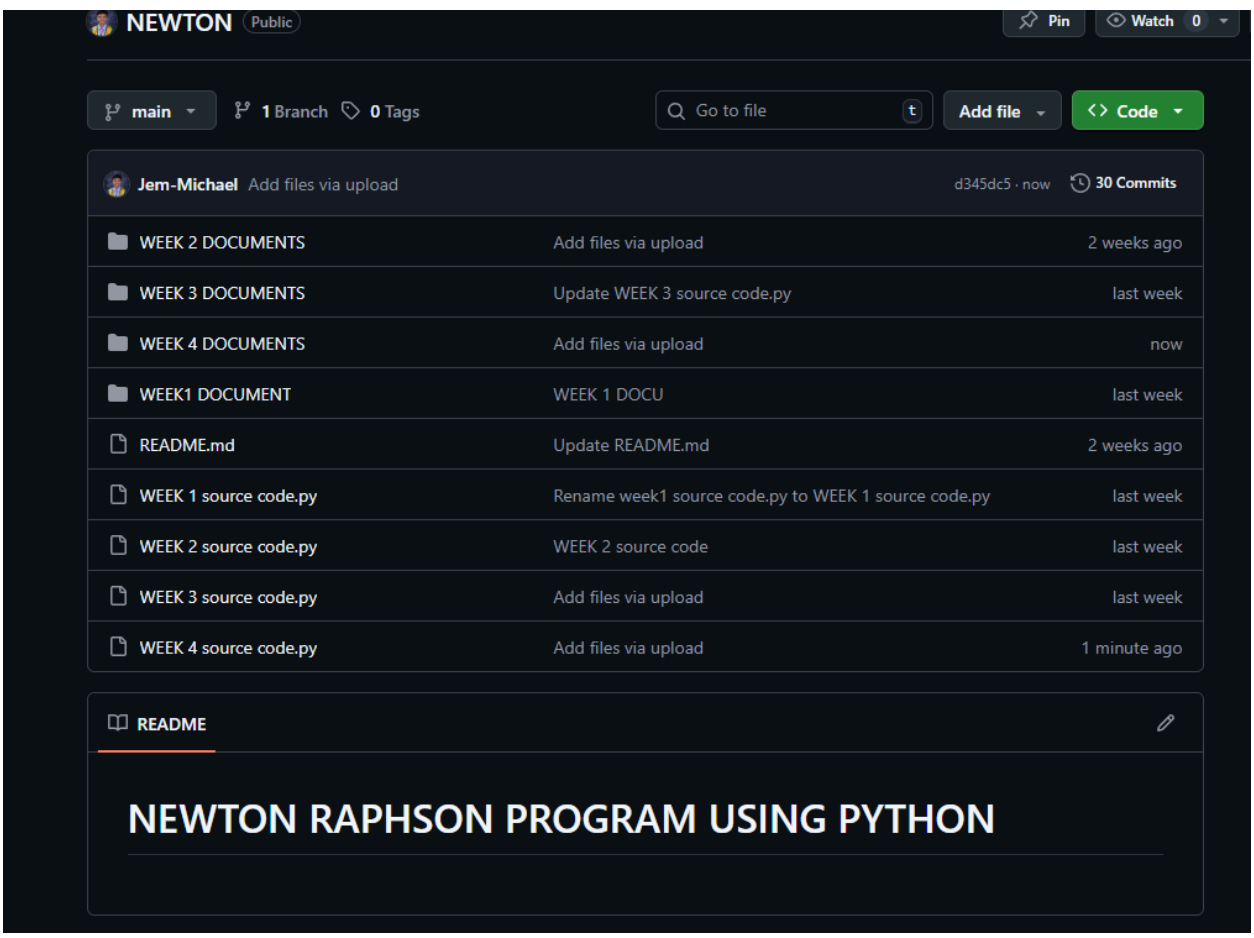
3	$x^3 - 3x + 1$	0	1.521380	1.521380	**Passed**

Required Outputs

- [x] ~~Solver works for at least 1 basic case (numeric or symbolic)~~
- [x] ~~At least 3 valid test cases documented (expected vs actual)~~
- [x] ~~Trail shows real computation steps (not only the final answer)~~

GITHUB REPO



Reflection

What part of the solver is still incomplete?

The system was able to use the Newton–Raphson method and show the full solution steps, there are still some things that can be improved. The system only uses the Newton–Raphson method and does not include other methods like the Bisection Method or False Position Method, which could help when the Newton–Raphson method does not work properly. Also, the solver does not show a graph of the function and its root, which could help users understand the process better. In the future, the system could be improved by adding graphs and better error messages when the method does not work or gives incorrect results.

Weekly Sign-off:

Group Leader Name/Signature: CHARLES RUSSEL O. SAN JUAN Date: 03/12/2026

Member Name/Signature: CHRISTINA GOMBA Date: 03/12/2026

Member Name/Signature: JOHN MICHAEL TULALIAN Date:03/12/2026